CURRICULUM VITAE

Joseph Bolton

j-sephb-lt-n.github.io/exploring_statistics/

PERSONAL DETAILS

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QUALIFICATIONS

Current part time study: **BSc(Hons) in Statistical Sciences**, Mathematical Statistics stream: Analytics, Operations Research (optimisation), Bayesian Statistics, Multivariate Analysis, Time Series, Statistical Computing, Biostatistics. University of Cape Town [UCT] (2019 - present)

BSc Mathematical Statistics: University of Cape Town [UCT] (2015-2017)

- Golden Key Member (top 15% of stream)

Performers' Diploma in Music – Jazz: University of Cape Town [UCT] (2010-2012)

- Dean's Merit List 2010, 2011, 2012
- Diploma Awarded with Distinction in Jazz Ensemble and Piano

Bachelor of Music in Classical Composition: University of Cape Town [UCT] (2006-2009)

- Dean's Merit List 2006, 2007, 2008
- Peter Klatzow Prize for Composition 2008

High School: Rondebosch Boys' High School (2001-2005)

- A-aggregate for matric (2005)
- Emilio Farella Performance Trophy (2005)
- Augusta Reynolds Award: Best Piano Sonata Entrant, Cape Town Eisteddfod (2004)

WORK EXPERIENCE

- Statistician/Data Scientist at Mr D Food (Apr 2020 - present)
    - Predictive Modelling:
        - Built customer churn prediction model: *XG-Boost* model trained on AWS
    - Python, Google BigQuery, ElasticSearch, SQL, AWS:
        - Automation of regular emails, data feeds, processes and reporting
    - Outlier Detection:
        - Automatic detection of poorly performing business units for early warning using *Lasso* regression model

- Statistician/Data Scientist at Eighty20 (Jul 2017 - Mar 2020)
  Content of previous projects:
    - Recommender Engines:
        - Built *content-based* recommendations engine based on raw text product descriptions of past item purchases, for individualised email product image recommendations (client: *The Foschini Group*)
        - Built *score-card* model, based on engineered attributes of past item purchases, for individualised email product image recommendations. Model in production for over 12 months. (client: *The Foschini Group*)
    - Time-series forecasting:
        - Order forecasting, as input to a voucher optimisation. Used *SARIMA* and *Exponential Smoothing* models. (client: *The Foschini Group*)
    - Automated customer selection through uplift modelling (modelling of heterogeneous treatment effects):
        - Used for customer selection in a recurring campaign. Used a *two-stage transformed outcome* model. In production. (client: *The Foschini Group*)
    - Mathematical optimisation:
        - Built optimisation model to allocate discount vouchers to customers across all African branches, maximising revenue and keeping cost below a prespecified threshold. Used a *tabu-search*-style model. Model performed above expectations. (client: *The Foschini Group*)
    - Predictive modelling:
        - Built model (in H2o) to predict customer propensity to shop online. Final model selected was a *Gradient Boosted Machine*. Model in production. (client: *The Foschini Group*)
    - Exploratory Data Analysis & Data Visualisation:
        - Presented brand basket analysis (using *association rules*) to client. (client: *The Foschini Group*)
        - Presented analysis of high-value customers to client (client: *The Foschini Group*)
    - Clustering:
        - Performed segmentation of high-value customers using *CLARA* algorithm (client: *The Foschini Group*)
    - R on-boarding (training) of new employees
    - Data wrangling in SQL (vertica).
    - Experimental design & hypothesis testing:
        - Design and monitoring of (ongoing) recurring direct marketing campaign (client: *The Foschini Group*)

    - Use of geo co-ordinates to enable unbiased survey sampling from informal settlements (including converting image data into geo-data).
- Internship at Eighty20 (2016-2017)
    - Text mining and REGEX
    - Data wrangling
    - Process automation
- Vacation internship at Arch Actuarial (2017)
    - Reviewing financial models
- Vacation internship at Momentum (2017)
    - Excel and VBA
- Author of the music chapters in *Study and Master: Creative Arts textbooks*, Grade 7, 8, 9 – published by Cambridge University Press; over 200,000 copies sold (2014-2018)
- Over-writing/editing music content in various Cambridge University Press publications (2012)
- Composer and Producer of Educational CD, *Study and Master: Creative Arts textbooks*, Grade 7, 8, 9 – published by Cambridge University Press (2012-2013)

SOFTWARE EXPERTISE

- R (Statistics) and R Shiny
- H2o Machine Learning
- Python
- AWS: S3, Athena, Sagemaker
- SQL (Vertica, MySQL, PostgreSQL, Google Bigquery, Athena)
- Microsoft Excel & VBA
- GIT
- Photoshop

PRIMARY STATISTICAL INTERESTS

- Uplift Modelling (estimation of heterogeneous treatment effects)
- Recommendation Engines
- Optimisation
- Predictive Modelling & Machine Learning
- Data Visualisation
- Frequentist and Bayesian Inferential Modelling
- Time series modelling and forecasting
- Deep learning
- Combinatorics

INTELLECTUAL COMPETENCIES

- Ability to analyse and synthesise; logical reasoning
- Problem-solving, mathematical/statistical skills
- Self-driven and self-teaching ability
- High-level networking skills
- High-level ability to interact with others
- Leadership
- Collaborative team-member
- Ability to innovate, conceptualise, visualise

REFERENCES

Colleague (statistician) at Eighty20:	Peter Smith 082 216 5509 peter.was.smith@gmail.com
Line manager at Eighty20:	Dino Bernicchi 072 213 2592 dinobernicchi@gmail.com
University statistics lecturer:	Şebnem Er 072 423 8876 sebnem.er@uct.ac.za